Procedural Generation of Solvable Levels in MARL Environment

Hugo Charels

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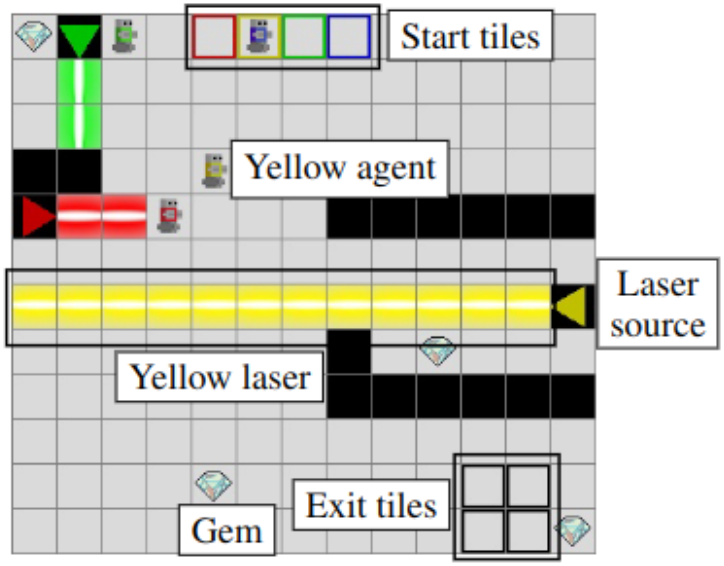


Context & Motivation

- Say what I do and why

LLE Solvability

- Say that the problem is harder than pathfinding
- Talk a bit about complexity
- Show options to make solver

Background: Satisfiability Problem

Definition (SAT)

Given a Boolean formula ϕ over variables $x_1, ..., x_n$, decide whether there exists an assignment $\alpha : \{x_1, ..., x_n\} \to \{0, 1\}$ such that $\phi(\alpha) = \top$.

Definition (CNF)

A formula is in conjunctive normal form if it is a conjunction of one or more clauses, where a clause is a disjunction of literals

$$F = \bigwedge_{i=1}^{m} \left(\bigvee_{j=1}^{k_i} l_{i,j} \right)$$

Theorem (Cook–Levin)

SAT is NP-complete.

Solver

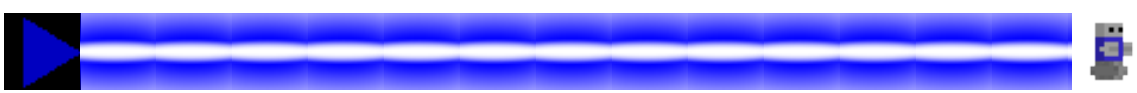
- Find a reduction (a way to transform a LLE solvability instance into a SAT instance)

Variables

- $a_{c,x,y,t}$: agent with color c at position (x, y) after t time steps
- $b_{c,d,x,y,t}$: beam of color c with direction d at position (x, y) after t time steps
- $l_{c,x,y,t}$: laser of color c active at position (x, y) after t time steps

Constraints

- Initializations : starting tiles and laser sources
- Movements : legal movements, unique position, no overlap and exit tiles
- Lasers : beam propagation



Agents movements

Legal movement:

$$\bigwedge_{c \in C} \bigwedge_{t \in T} \bigwedge_{(x,y) \in P} a_{c,x,y,t} \rightarrow \bigvee_{(x',y') \in \text{next}(x,y)} a_{c,x',y',t+1}$$

Agents movements

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At most one agent per tile:

$$\bigwedge_{c \in C} \bigwedge_{t \in T} \bigwedge_{\substack{(x_1,y_1),\\(x_2,y_2) \in P\\(x_1,y_1) \neq (x_2,y_2)}} \neg a_{c,x_1,y_1,t} \vee \neg a_{c,x_2,y_2,t}$$

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Global

At most one agent per tile:

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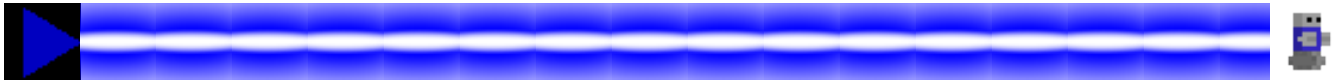
Local

Agent should come from a legal movement:

$$\bigwedge_{c\in C}\bigwedge_{t\in T}\bigwedge_{(x,y)\in P} a_{c,x,y,t+1} \rightarrow \bigvee_{(x',y')\in \text{next}(x,y)} a_{c,x',y',t}$$

At most one future position:

$$\bigwedge_{c\in C}\bigwedge_{t\in T}\bigwedge_{\substack{(x',y'),(x'',y'')\in \text{next}(x,y)\\(x',y')\neq(x'',y'')}} \neg a_{c,x',y',t} \vee \neg a_{c,x'',y'',t}$$



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At most one agent per tile:

$$\bigwedge_{c\in C}\bigwedge_{t\in T}\bigwedge_{\substack{(x_1,y_1),\\ (x_2,y_2)\in P\\ (x_1,y_1)\neq(x_2,y_2)}} \neg a_{c,x_1,y_1,t} \vee \neg a_{c,x_2,y_2,t}$$

$$f(n) = \binom{n}{2} = \frac{(n-1)n}{2} \in O(n^2)$$

Local

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Global

At most one agent per tile:

$$\bigwedge_{c \in C} \bigwedge_{t \in T} \bigwedge_{\substack{(x_1,y_1), \\ (x_2,y_2) \in P \\ (x_1,y_1) \neq (x_2,y_2)}} \neg a_{c,x_1,y_1,t} \vee \neg a_{c,x_2,y_2,t}$$

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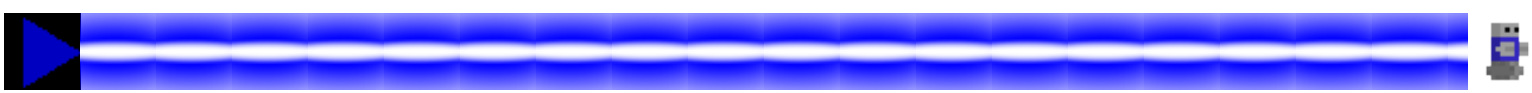
$$g(n) = (f(5) + 1)n = 11n \in O(n)$$



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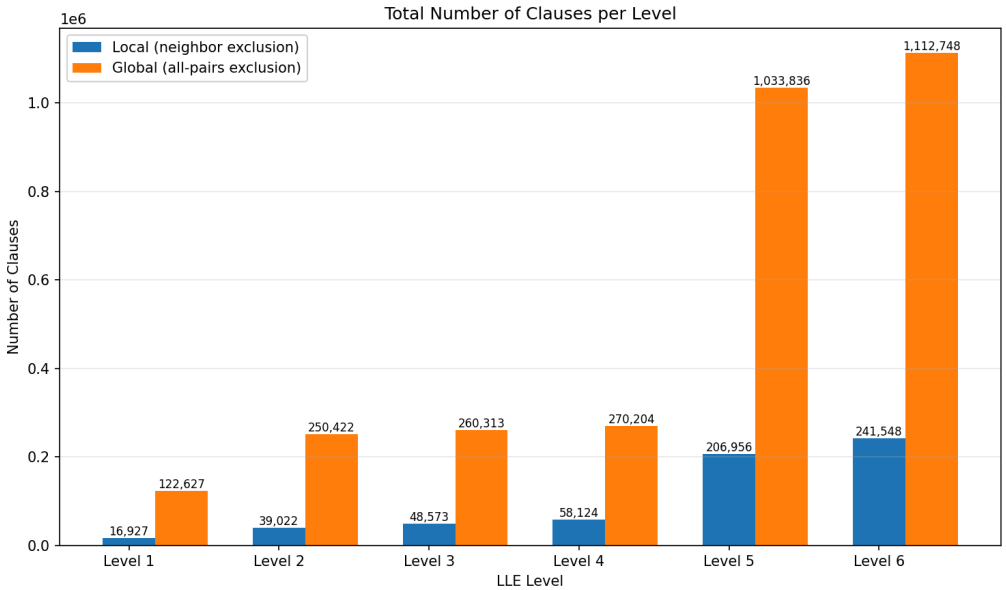
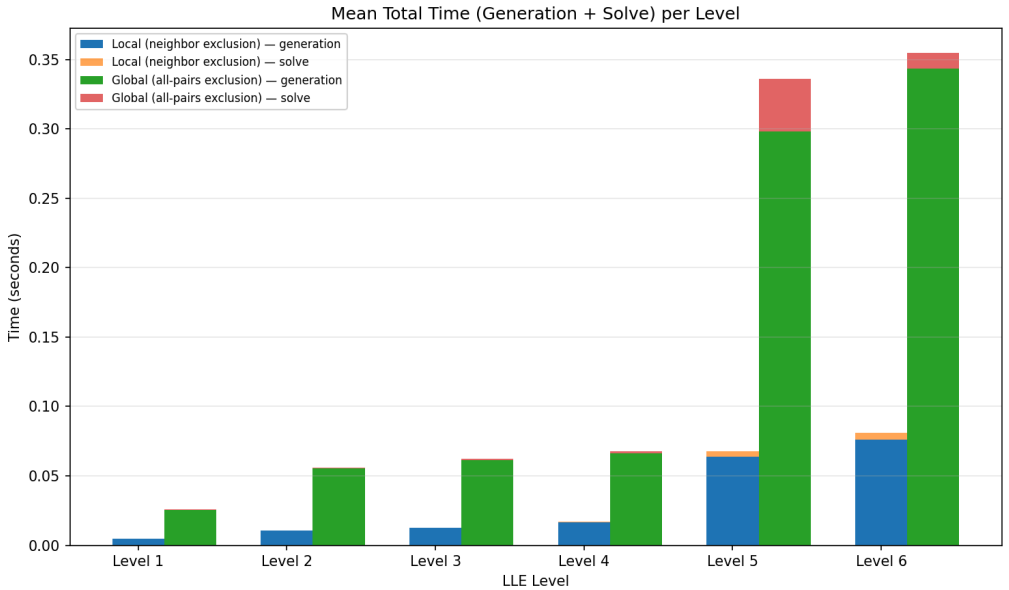
$$g(n) = (f(5)+1)n = 11n \in O(n)$$

$f(n) < g(n)$ when $n < 23$ $\quad$ $f(n) = g(n)$ when $n = 23$ $\quad$ $f(n) > g(n)$ when $n > 23$

Plots difference





Future work

End

Outline